AKADEMIKA

ACL-NP

NOISE & AUDIO AMPLIFIER KIT

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ANALOG COMMUNICATION is one of the important subject need to teach while learning electronics
Whether fiber optic system are for transmitting voice, video & audio or training regardless of application, it is our vision to provide you with the product you need, ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training product. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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Maharashtra- 411038

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

Set all accessories in order on unpacking and check integrity with respect to the packing list. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **ACL-NP kit**, signals are to be monitored with an oscilloscope, the scope input channel should be ac coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit/System has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

INDEX

TECHNICAL SPECIFICATIONS

Clock Generator
Frequency : 2MHz

Onboard Signal
SINE WAVE
Frequency : 1 to 10 KHz.
Amplitude : 0 to 2Vpp.

Noise Generator
Pseudo Random Noise Source
Number of Bits : 32 bit.
Output Amplitude : 0 to 1V.
Noise Bandwidth **:** 2 MHz.

Signal Attenuation an Adder
Adjustable from 0 to the maximum of input value
Signal + Noise Adder Stage

Low Pass Filter
Cut Off Frequency : 3.4 KHz.

Power Meter and Display
Input signal Amplitude : 0 to 2Vpp.
Timer : 1 to 15 seconds.
Display : 2 Digits Seven Segment

Switch Faults
: 4 Switch Faults are provided on board to study different effects on circuit.

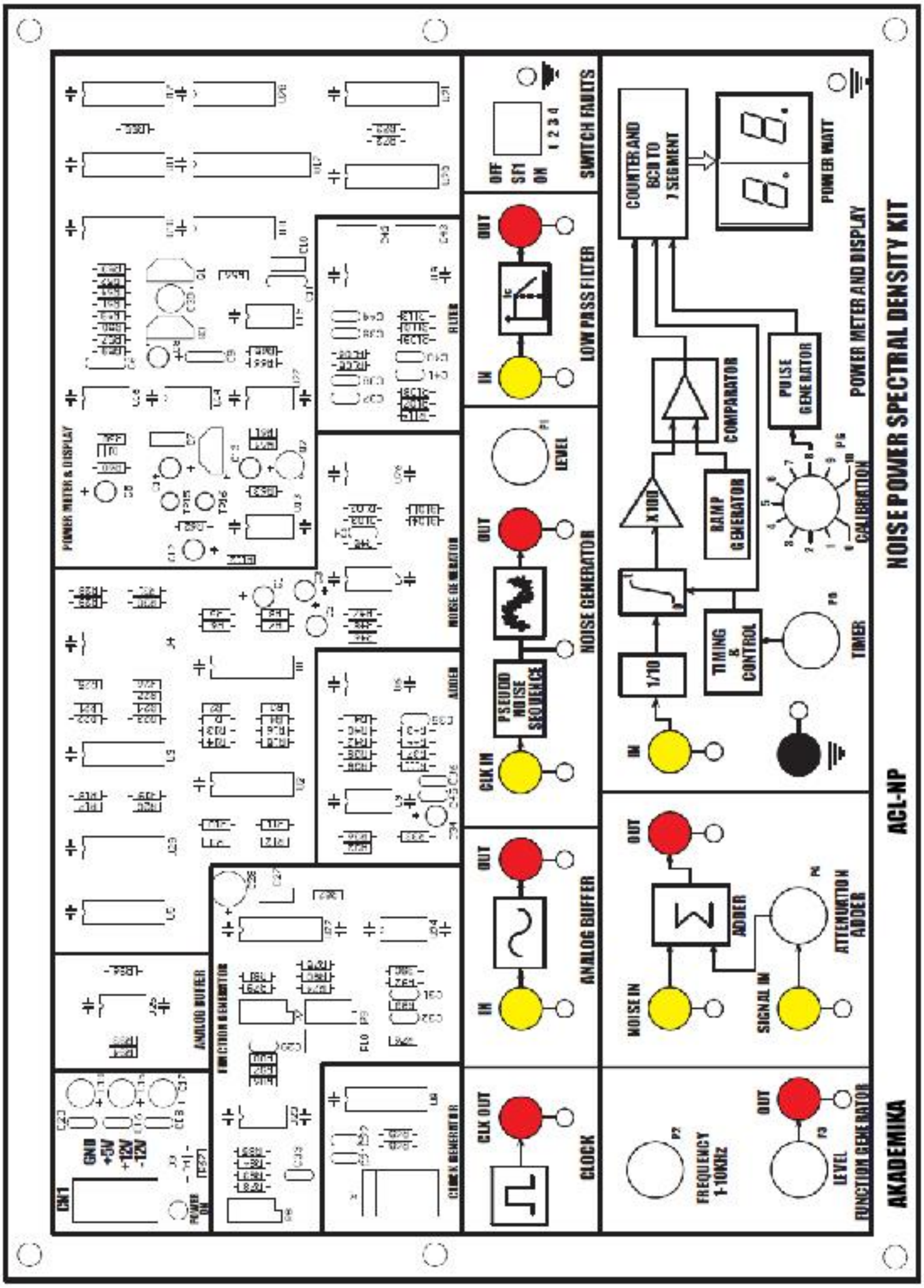
Interconnection
: 2 mm Banana Socket

Accessories:

	ACL-NP	Quantity
1	Mini Links PCS2 Patch cord Male To Male	7
2	ACL-NP EXPT Manual	1
3	Power Supply	1
4	Power cord	1

NOTE: Power Meter Counts For First 15 Seconds Then Shows Stable Output For 15 Seconds, This Cycle Repeats Continuously. All Measurements Should Be Done When Display Is Stable.

FUNCTIONAL BLOCK



INTRODUCTION

The series **ACL-NP, NOISE POWER SPECTRAL DENSITY KIT** is a modular system for the development of exercises and theoretical-experimental courses to understand the noise effect in Analog Circuits and in AM and FM Communication Systems.

The unit consists of a set of modules, each including one or more functional blocks typical of communication systems.

The system is highly innovative from a technological as well as an educational point of view. The modules are used as "basic blocks" to build up in a flexible way the different communication systems, and to examine all the peculiar operating characteristics.

The only external instruments required are Microphone, Speaker, Power Supply Unit and an Oscilloscope. This Kit can be interfaced with ACT Series and Analog Communication Systems.

Utmost care has been laid in the design and quality control of all circuits, to ensure the repeatability of the results of the experiments.

ACL-NP

Function Generator

The integrated circuit U22 (ICL8038) generates sine waveform (pin 2). The capacitors which are connected to pin 10 enable to select the range (1 KHz or 10 KHz max). The frequency is varied with the pot **P2 FREQ**. (RV5 limits the inferior frequency). The Pots P7, P8, P9 and P10 adjust the symmetry and the form of the sine signal. The pot **P3 LEVEL** varies the signal amplitude.

Analog Buffer

This is a unity gain amplifier built around op-amp U25 (IC 3140).

Noise Generator

The noise generator is a pseudo random generator. The shift generators U1, U2, U3, U4 (74C164) are used to generate the 32 bit pn sequence. The pn sequences with delays are sent through an adder circuit U6 (CA3140) to generate the noise signal. It generates a noise of about 1V rms. The potentiometer at the output of the adder is used to vary the noise power. It generates noise to a bandwidth of 2 MHz.

Clock Generator

The 2.048 MHz crystal Oscillator generates a 2.048MHz clock.

4^{th} Order Low Pass Filter

4^{th} order Butter worth Low Pass Filter is built around U26 (IC LF353) which filters out the noise in the signal.

Power Meter and Display

The power meter consists of a multiplier, followed by an integrator and dump filter and an ADC and seven segment display section.

Multiplier

The multiplier U13 is a four quadrant analog multiplier with an internal attenuation factor of 10. If both the inputs are same, then it functions as a squarer.

Integrate and Dump Filter

An integrate and dump filter is a device that integrates the incoming signal for a given period of time then dumps the accumulated value when required and starts integrating from zero again.
This section consists of a normal integrator circuit, a sample and hold circuit, and timing circuits so that they may operate in conjunction as an integrator and dump filter.
The integrator is a simple LPF (combination of 3.9K and 100muF). The U16 (555) in the astable mode (with 15k and 270 K) is the controller for timing. This 555 produces pulse trains of 1s on time and 29s off time. The transistor BC107 turns on during the on time of 555 outputs. Hence the capacitor discharges to ground whatever it has integrated till then. Now the 555 turns off after 1s and stays so for the next 29s. When the NE555 (U16) turns off its falling edge triggers the monostable multivibrator NE555 (U14) which has a pot. This monostable output stays high for a period determined by the pot position and can be varied from 0 to 15s.This gives the period of integration.

ADC and Display Section

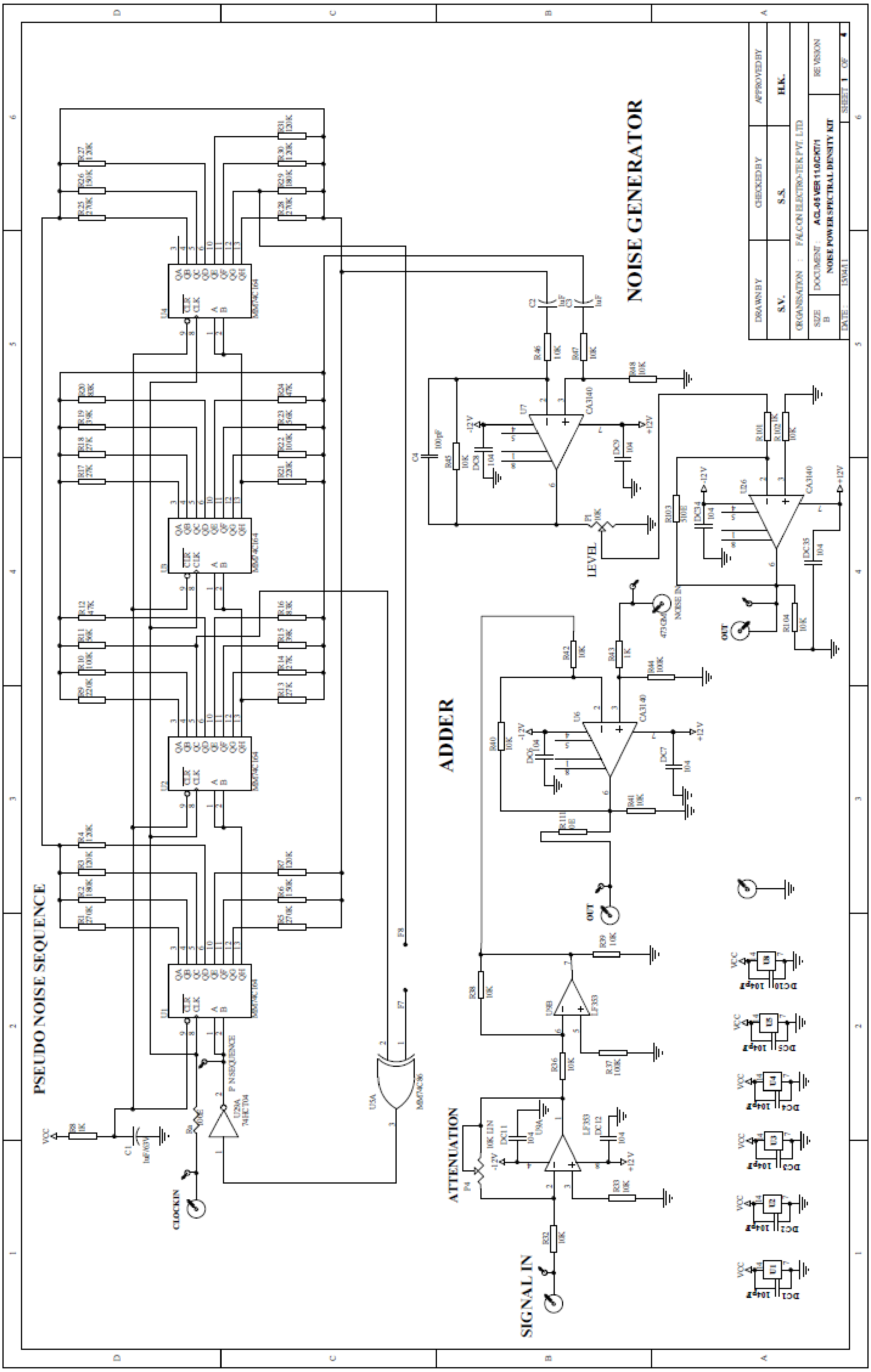
The (ADC) U10 consists of three comparators com 1, 2, 3 and a ramp generator (the BC557 section) and a pulse generator astable (555) U15 section of 10nF and 100Kpot).
The ramp generator produces a ramp waveform of peak value 12V. The comp 3 compares the boosted dc output of integrate and dump, with the ramp waveform. Whenever the dc value is higher it produces a high output. This comp3 output serves to gate the clock pulse produced by astable 555 to the counter U18 (74C90). The principle of operation is that when the output power is 0.1W the boosted value at comp3 input is 1W (because of an excessive gain of 10). Now the comp3 o/p stays high for a particular interval when the ramp is smaller than 1. During this interval the pulse generator generates 10 pulses. So counter counts 10. This is shown in the display as 10. This has to be interpreted as 0.10. The main thing is the adjustment of pot so that just 10 pulses are produced in the interval when comp 3 remains high. To achieve this pot is provided for accurate calibration. It is advisable to calibrate the ADC regularly with a known DC input.
The purpose of com1 and com2 is to latch the counter value to the 7 segment converter U20 & U21 (74C47) and to reset the counter respectively. The com1 o/p goes high when the ramp reaches its peak. So whatever value is counted in one cycle is latched to (74C47). When the ramp discharges and just starts a new cycle com2 resets the counter. Thus the ADC operation is repeatedly performed with each ramp cycle.

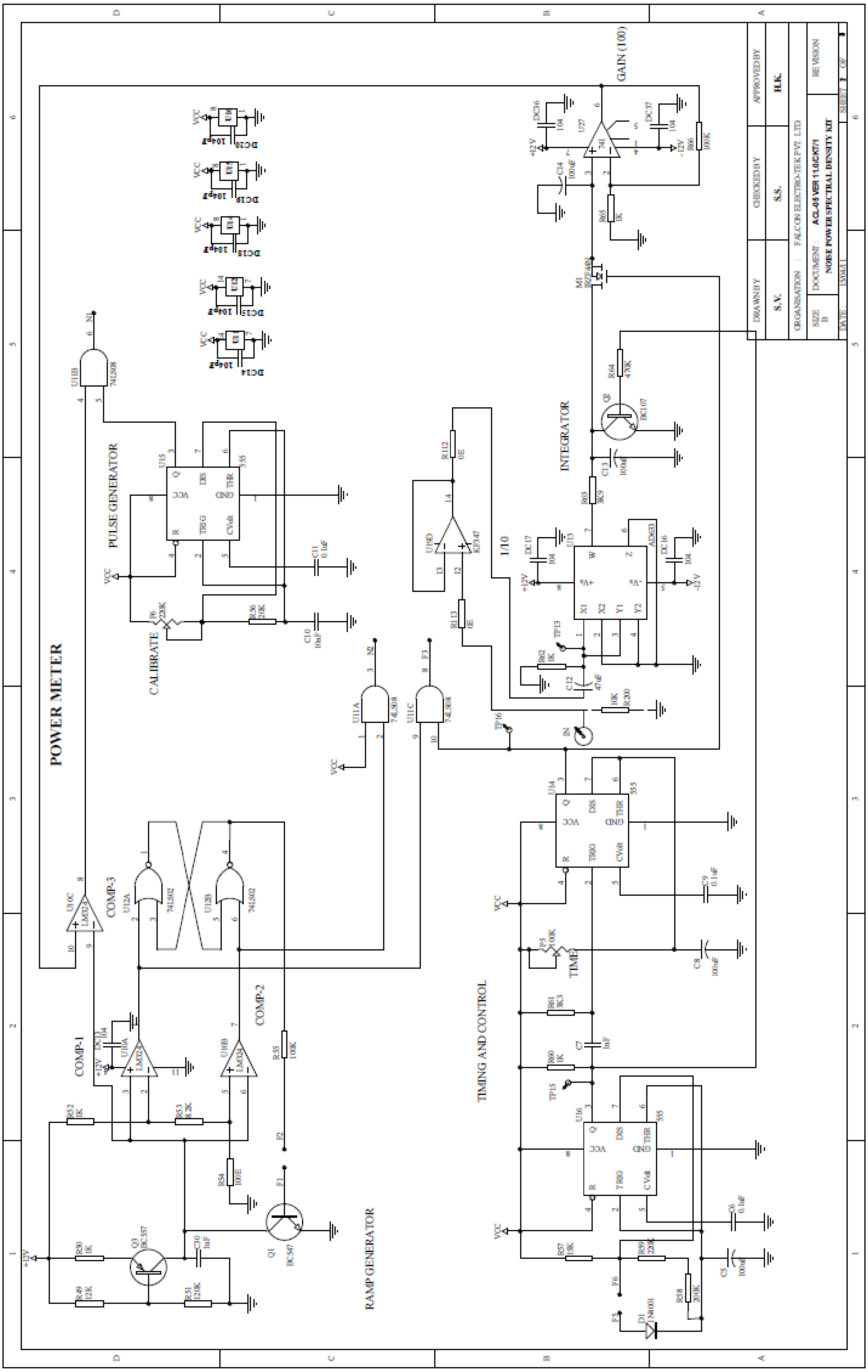
SWITCH FAULTS

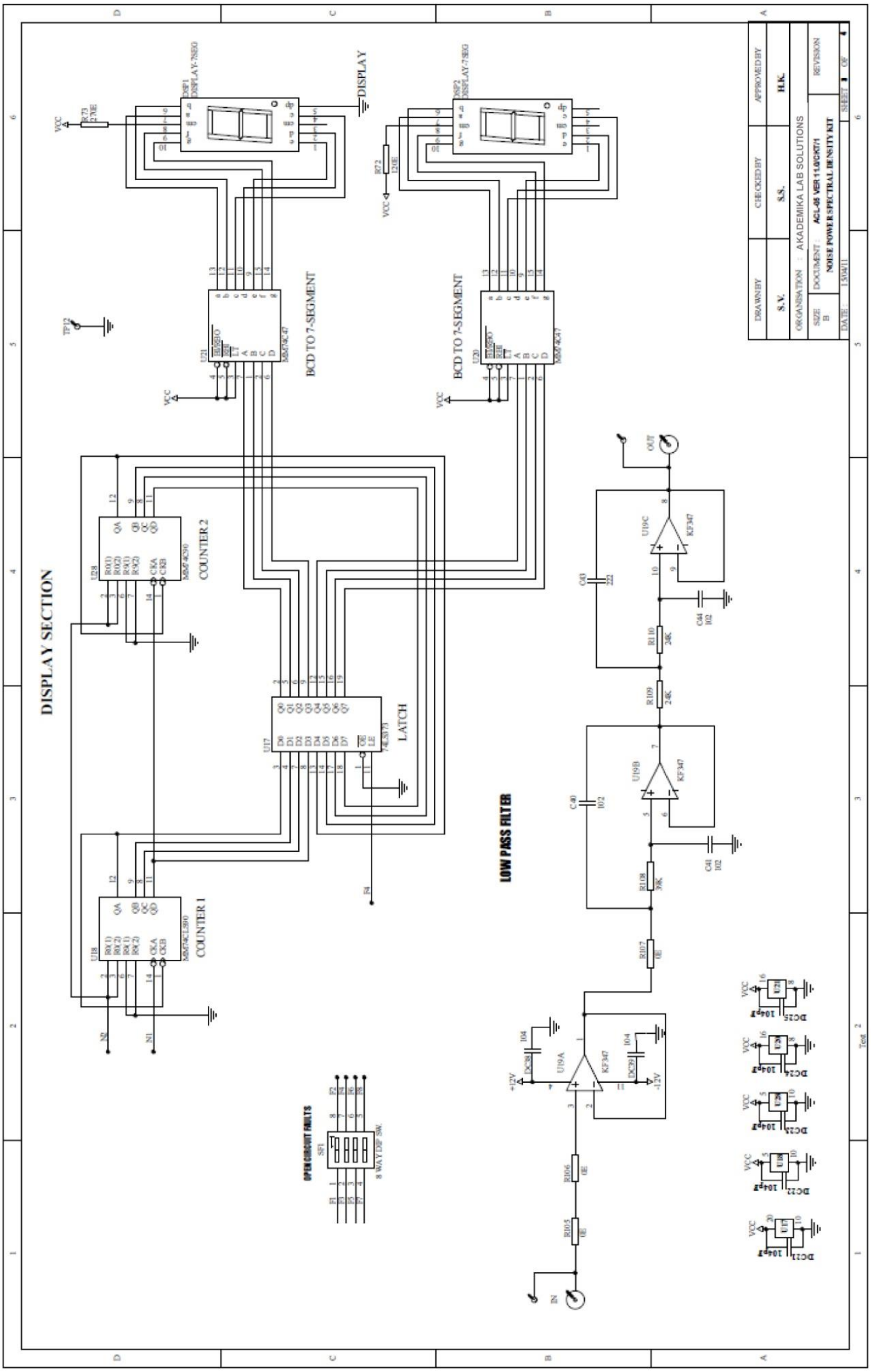
Switch fault section on **ACL-NP** is provided with **4** switches. These switches can be used to simulate fault conditions in various parts of the circuit.

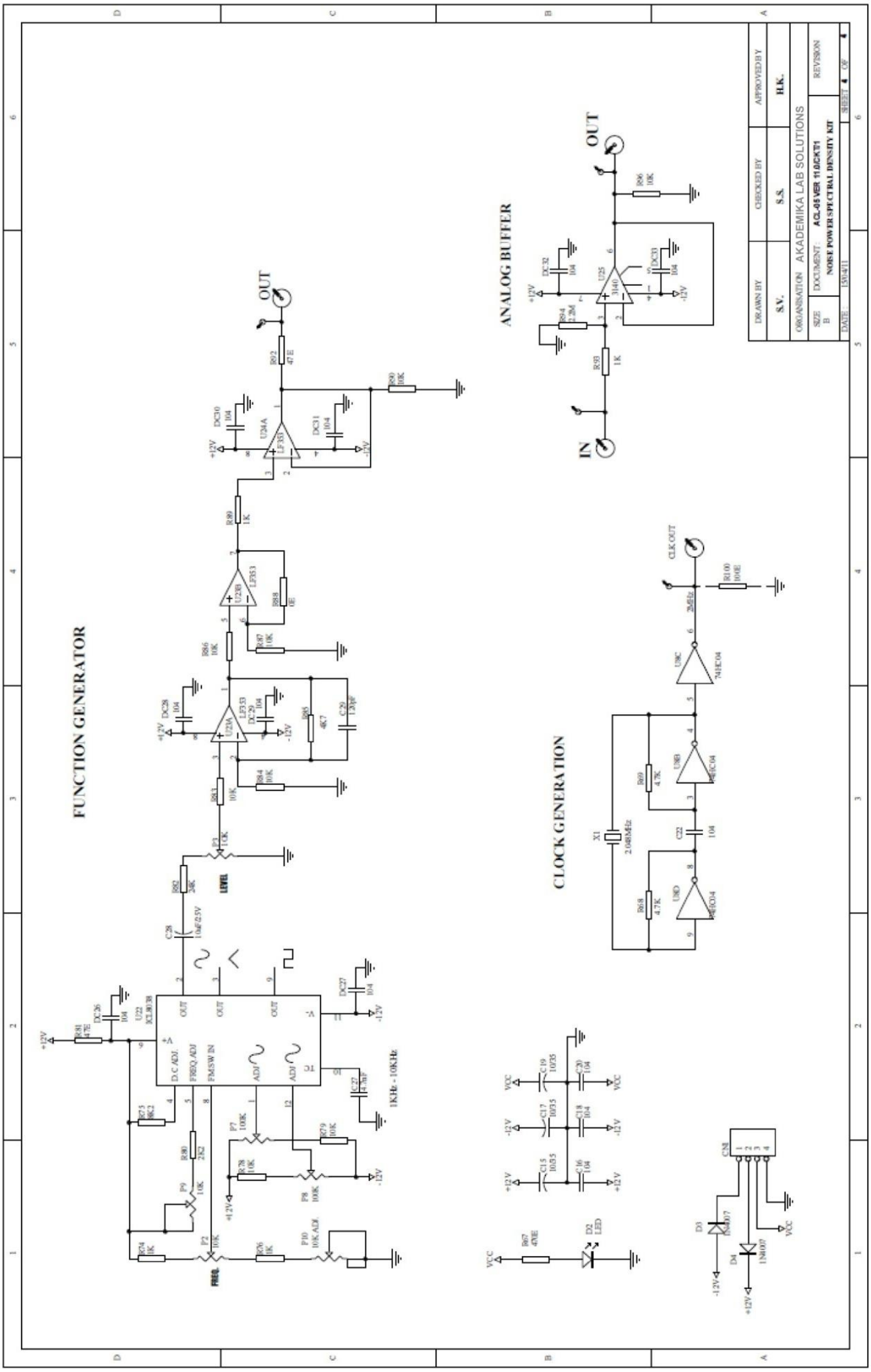
- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will open diode D1 across R58 & R59; this will increase ON time of astable multivibrator to about 52 seconds. Thus, time between successive readings get increases.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will open latch enable signal to U17 (pin11) from U11 (pin 8) the value controlled by counter won't get latched.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will open connection between transistor Q1 base & R55. So ramp generation will be lost. This will disable the counter and ADC operation will be affected.

CIRCUIT DIAGRAM




POWER METER
PULSE GENERATOR
CALIBRATE
INTEGRATOR
GAIN (100)
1/10
TIMING AND CONTROL
TIME
RAMP GENERATOR
COMP-1
COMP-2
COMP-3
DRAWN BY
S.V.
CHECKED BY
S.S.
APPROVED BY
H.K.
ORGANISATION : FALCON ELECTRO-TEK PVT. LTD
DOCUMENT : ACL-05 VER 11.0/CKT/1
NOISE POWER SPECTRAL DENSITY KIT
REVISION


DISPLAY SECTION
DISPLAY
BCD TO 7-SEGMENT
BCD TO 7-SEGMENT
COUNTER 2
COUNTER 1
LATCH
LOW PASS FILTER
AKADEMIKA LAB SOLUTIONS
NOISE POWER SPECTRAL DENSITY KIT


FUNCTION GENERATOR
ANALOG BUFFER
CLOCK GENERATION
AKADEMIKA LAB SOLUTIONS
NOISE POWER SPECTRAL DENSITY KIT

EXPERIMENT NO.1

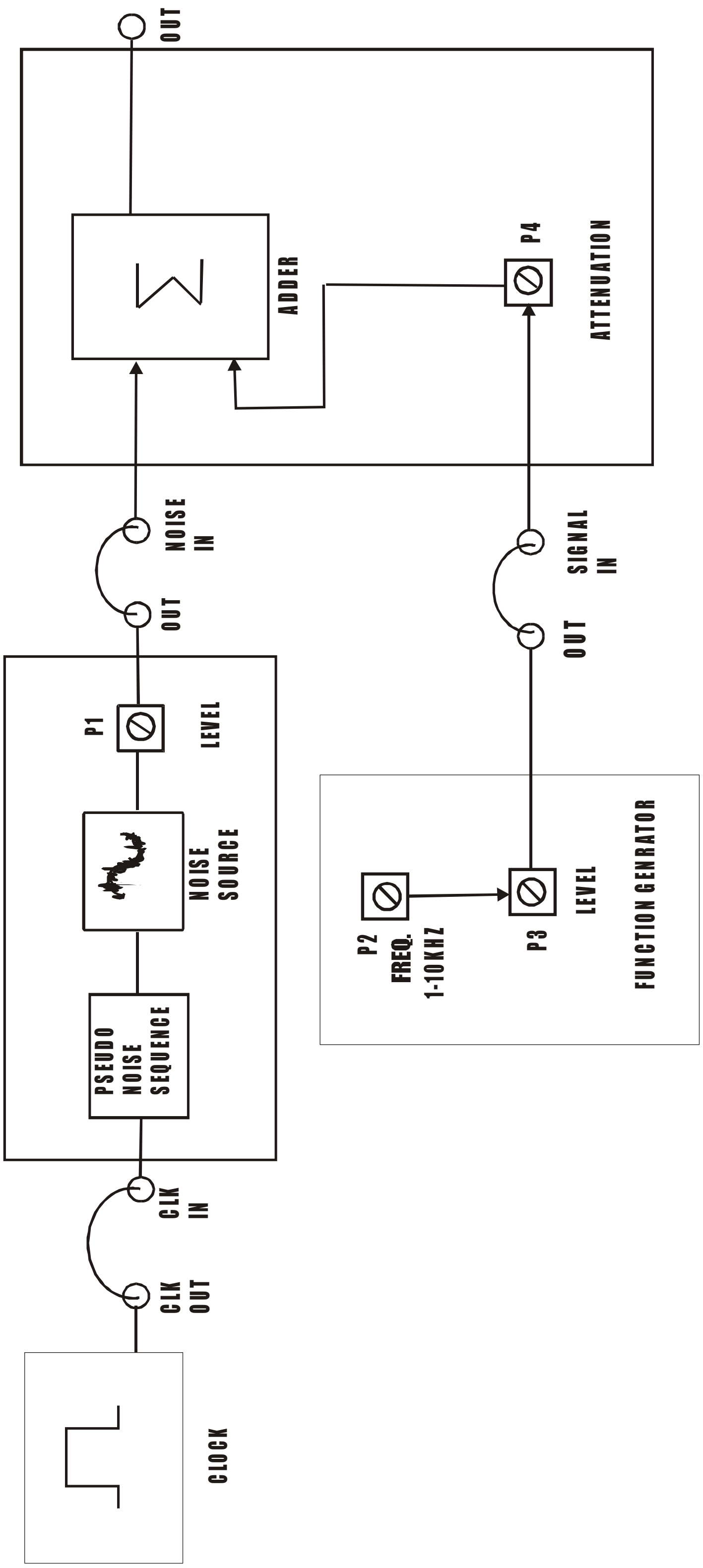


FIG.NO.1.1

BLOCK DIAGRAM TO OBSERVE THE EFFECT OF NOISE ON VARIOUS ANALOG SYSTEMS.

EXPERIMENT NO.1

NAME: -

To Study Effect of Noise on Various Analog Systems.

OBJECTIVES:-

A) To examine the operation of a noise generator.
B) To examine the operation of a signal attenuation network.

THEORY: -

We are all familiar with the Noise and its ill effects not only on human health but also on electronic equipment. Apparently, `Noise' is an undesirable electrical disturbance in a useful frequency band.
There are several sources of noise encountered in electronics and in everyday life. Some of the noise sources are: Howls, hum, ambient noise, back-ground noise, contact noise, circuit noise, component noise, carrier-wave noise, cosmic noise, flicker noise, line noise, modulation noise, random noise, radio receiver noise, shot noise, valve noise, transistor noise, thermal noise, diode noise, diode noise, pink noise, and white noise.
In electronic circuits (AF or RF) noise in the first and second stage must be kept extremely low otherwise the signal to noise ratio will become high and the incoming signal will be embedded in the pool of noise. We try to reduce noise by various means but the residual noise can't be removed fully.
Noise source has become a very useful item in electronics. Noise source is used for adjusting VHF & UHF radio receivers. It is also essential for the synthesis of many sounds. These range from musical instruments such as pipe organ and percussive devices. We can now synthesize natural sounds like wind, rain and surf. In the field of man-made sounds such as steam engines, explosions and gunfire can be easily synthesized by noise.

What is Noise?
Noise is caused by highly complicated physical and thermodynamic processes. Briefly, it is the random movement of electrical charge carriers. Noise increases with rise in temperature. At absolute zero (-273°C=0K°) noise is zero because at this temperature all movements of charge carriers are frozen, and there is no trace of noise.

Generation of Noise
The noise generator is a PN sequence generator. PN sequence stands for pseudo-random noise binary sequence. PN sequence generator consists of shift register network. D flip-flops are used as shift register. D flip-flops are arranged in such a way that data input except first flip-flop input is the output of preceding flip-

flop. The input of first flip-flop is the output of a parity generator, generally constructed using an array of Ex-OR logic gates. A parity generator generates an output which is at logic 0 when an even number of inputs are at logic 1 and generates an output which is at logic 1 when an odd number of inputs are at logic 1. The parity generator inputs are outputs of selective flip-flops. The character of PN sequence generated depends on the number N of flip-flops used and on the selection of which flip-flop outputs are connected to the parity generator.
The outputs of Shift Registers are tapped through Register network to generate amplitude noise signal, which is about 1 Vrms.

EQUIPMENTS: -

- ACL-NP.
- Power supply.
- 20MHz Dual trace oscilloscope.

NOTE: Keep all switch faults in OFF position.

A) Examine the Operation of A Noise Generator.

PROCEDURE:-

- Connect the power supply with proper polarity to the kit **ACL-NP** and switch it **ON**.
- Ref to the fig. **1.1** & carry out the following connections and settings.
- Connect the **CLK OUT** post to **CLK IN** post of noise generator.
- Connect **OUT** post of noise generator to **NOISE IN** post of adder.
- Keep the **LEVEL** pot **P1** of noise generator to minimum position.
- Connect **OUT** post of function generator to **SIGNAL IN** post of adder.
- Keep the attenuation pot **P4** to minimum position.
- Keep freq. pot **P2** to minimum position i.e. at **1 KHz.**
- Using pot **P3** Keep signal amplitude at **1Vpp**.
- Observe the signal at **OUT** post of adder.
- Increase the level of noise by varying the **LEVEL** pot **P1** of noise generator.
- Observe the signal at **OUT** post of adder. You will observe that the noise get superimposed on the original signal.

B) Examine the Operation of a Signal Attenuation Network

PROCEDURE:-

- Repeat the procedure from step no. 1 to step no.10 above.
- Now keep the attenuation at maximum position by varying pot **P4.**
- Observe the signal at **OUT** post of adder and check that the output varies from a maximum, which is about the input amplitude to a minimum of 0 Volt.
- Vary the input signal frequency and check that the channel presents an almost flat response.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.

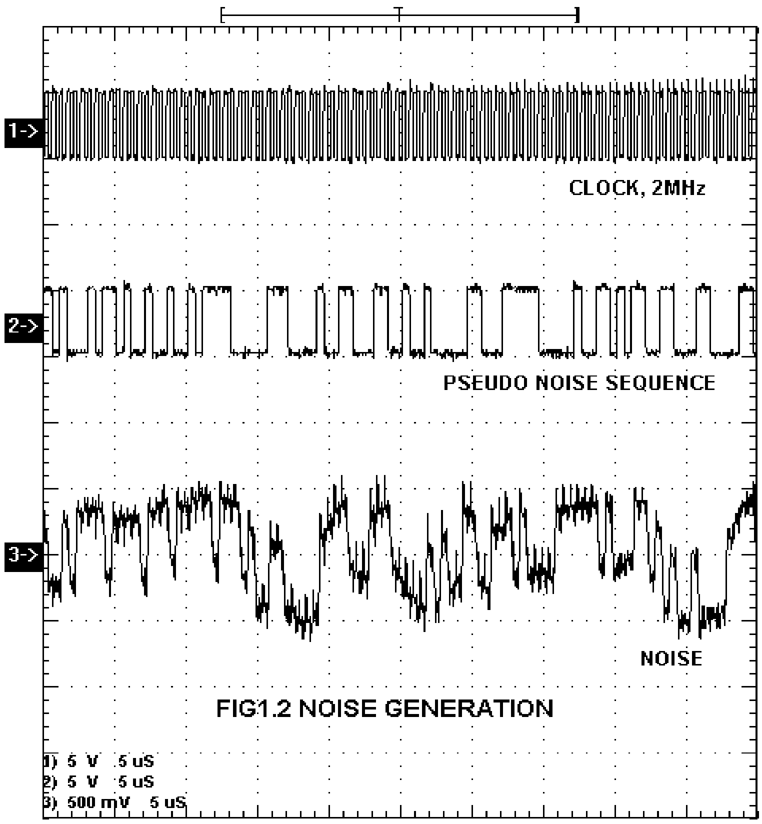


FIG1.2 NOISE GENERATION

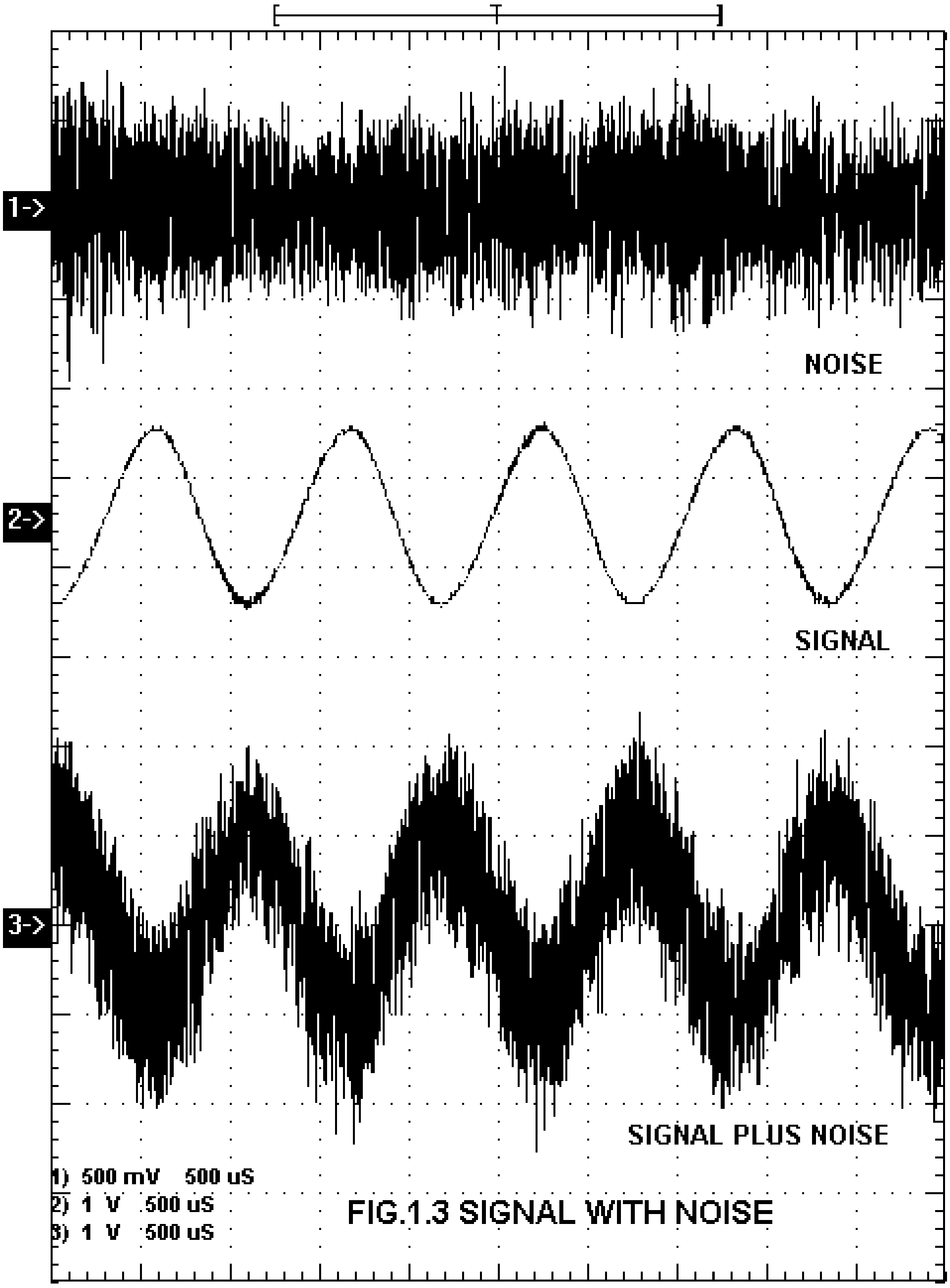


FIG.1.3 SIGNAL WITH NOISE

EXPERIMENT NO.2

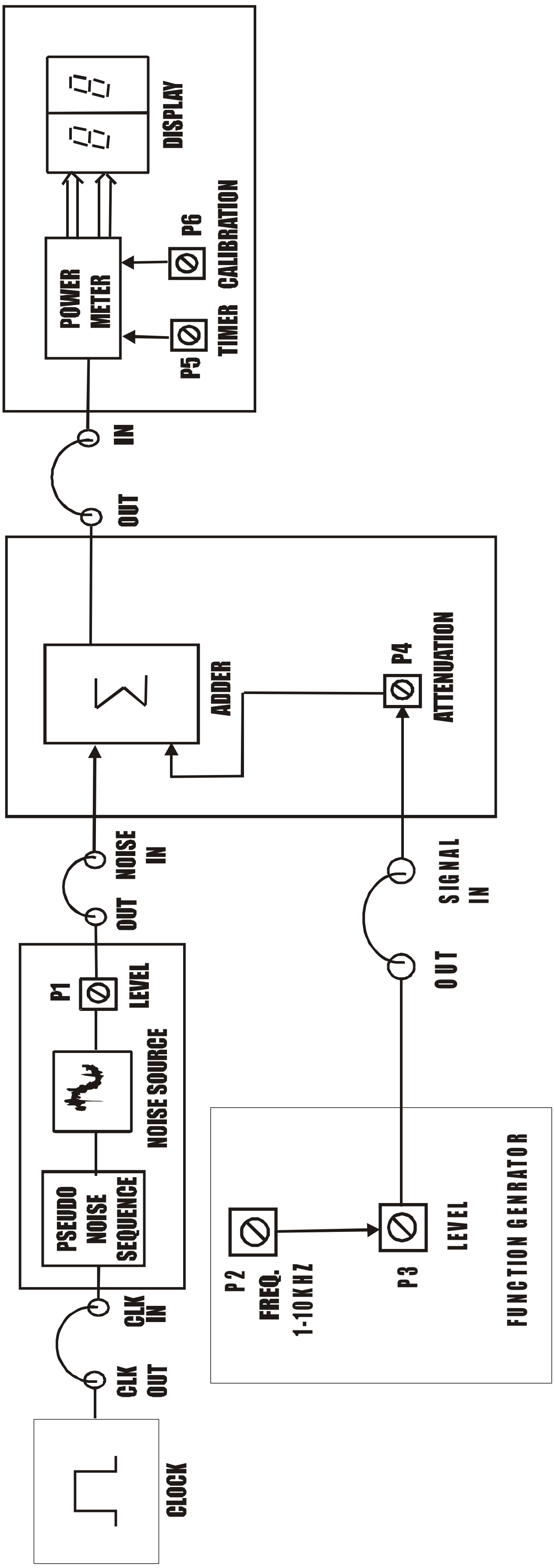


FIG. NO 2.1

CALCULATE THE SIGNAL TO NOISE RATIO.

EXPERIMENT NO.2

NAME:-

To Study Calculation of the Signal to Noise Ratio.

OBJECTIVES: -

Measurement of signal to noise ratio.

THEORY:-

Signal- To- Noise Ratio

The calculation of the equivalent noise resistance of an amplifier, receiver or device may have one of two purposes or sometimes both. The first purpose is comparison of two kinds of equipment in evaluating their performance. The second is comparison of noise and signal at the same point. Therefore same point to ensure that the noise is not excessive. In the second instance, and also when equivalent noise resistance is difficult to obtain, the signal-to-noise ratio S/N is very often used. It is defined as the ratio of signal power to noise power. Signal to noise ratio is calculated using diff procedures for diff modulation techniques.

SNR = signal power / noise power

EQUIPMENTS: -

- ACL-NP.
- Power supply.
- 20MHz Dual trace oscilloscope.
- Amplitude Modulator circuit.

NOTE: Keep all switch faults in OFF position.

PROCEDURE: -

- Connect the power supply with proper polarity to the kit **ACL-NP** and switch it **ON**.
- Ref to the fig. **2.1** & carry out the following connections and settings.
- Connect the **CLK OUT** post to **CLK IN** post of noise generator.
- Connect **OUT** post of noise generator to **NOISE IN** post of adder.
- Keep the **LEVEL** pot **P1** of noise generator to minimum position.
- Connect **OUT** post of function generator to **SIGNAL IN** post of adder.

- Keep freq. pot **P2** to minimum position i.e. at **1 KHz.**
- Using pot **P3** Keep signal amplitude at **2Vpp**.
- Connect **OUT** post of adder to **IN** post of power meter.
- Keep Timer pot **P5** to maximum position.
- Keep the signal attenuation at minimum using pot **P4**.
- Vary the calibration pot **P6** in such a way that you should get power output approximates to the theoretical value of the power of the signal (i.e. Keep between 4 & 5).
- Increase the noise level by using **LEVEL** pot **P1.**
- Keep the attenuation pot **P4** to minimum position.
- Measure noise power **Pn** on power meter.
- Repeat the procedure by increasing the noise level in steps.
- Calculate signal to noise ratio by using formula

SNR = (Ps / Pn)

- Similar observations and calculations can also be performed by using AM signal as input signal.
- Vary the calibration pot **P6** in such a way that you should get power output, which approximates to the theoretical value of the power of the signal. (i.e. Keep between 7 & 8)

Sr. No.	**Psn**	**Pn**	**Ps = (Psn – Pn)**	**SNR**
1.	2.1	0.1	2.0	20
2.	2.2	0.2	2.0	10
3.	2.4	0.3	2.1	7
4.	2.5	0.4	2.1	5.25
5.	2.6	0.5	2.1	4.2

(Table given for reference only actual values may differ from above)

Theoretical Power Calculations

A. Sine Wave

Signal Power $= E^2/2$

$= (Vm\ Sinwt)^2/2$

$= Vm^2\ Sin^2wt/2$

$= Vm^2/2 \times (1\text{-}Cos2wt)/2$

$= Vm^2/2$

$= 2^2/2$

$=$ 2 Watt.

B. AM Signal

Let Vc (Carrier Amplitude) = 1.8Vpp

Let Vm (Modulating Signal Amplitude) = 0.5Vpp

Therefore Pc (Power of Carrier signal) = $Vc^2/2$ = 1.62

m = Vm / Vc

= 0.5 / 1.8

= 0.28

Total Power (Pt) $= Pc\ (1 + m^2/2)$

$= 1.62 \times (1 + (0.27)^2/2)$

= 1.68W

NOTE: Power meter counts for first 15 seconds then shows stable output for 15 seconds, this cycle repeats continuously. All measurements should be done when display is stable.

NOTE: Make Sure PRBS is generate.

SWITCH FAULTS: -

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will open diode D1 across R58 & R59. This will increase ON time of astable multivibrator to about 52 seconds. Thus, time between successive readings get increases.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will open latch enable signal to U17 (pin11) from U11 (pin 8) the value controlled by counter won't get latched.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will open connection between transistor Q1 base & R55. So ramp generation will be lost. This will disable the counter and ADC operation will be affected.

EXPERIMENT NO.3

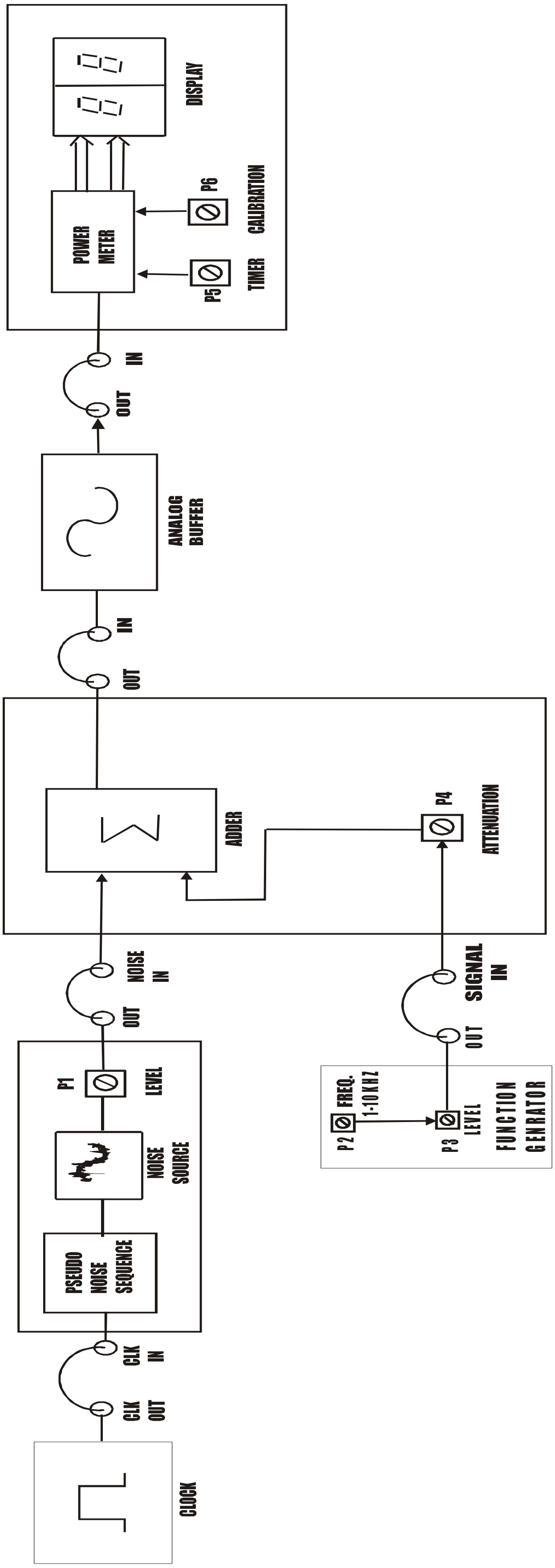


FIG. NO 3.1
FOR MEASUREMENT OF NOISE FIGURE

EXPERIMENT NO.3

NAME: -

To Study Measurement of Noise Figure.

OBJECTIVES: -

Measurement of noise figure of Amplitude Modulator & Demodulator circuit.

THEORY:-

Noise Figure
The noise figure F is defined as the ratio of the signal-to-noise power at the input to the signal-to-noise power at the output of the circuit stage.

F = (SNR) i/p / (SNR) o/p.

EQUIPMENTS:-

- ACL-AM,ACL-AD, ACL-NP
- Power supply.
- 20MHz Dual trace oscilloscope.
- Amplitude Modulator & Demodulator circuit.

NOTE: Keep all switch faults in OFF position.

PROCEDURE:-

A. Input Signal: Sine Wave

- Connect the power supply with proper polarity to the kit **ACL-NP** and switch it **ON**.
- Ref to the fig. **3.1** & carry out the following connections and settings.
- Connect the **CLK OUT** post to **CLK IN** post of noise generator.
- Connect **OUT** post of noise generator to **NOISE IN** post of adder.
- Keep the **LEVEL** pot **P1** of noise generator to minimum position.
- Connect **OUT** post of function generator to **SIGNAL IN** post of adder.
- Keep freq. pot **P2** to minimum position i.e. at **1KHz.**
- Using pot **P3** Keep signal amplitude at **2Vpp**.
- Connect **OUT** post of adder to **IN** post of Analog Buffer.
- Connect **OUT** post of Analog Buffer to **IN** post of power meter.
- Keep Timer pot **P5** to maximum position.

- Vary the calibration pot **P6** in such a way that you should get power output approximates to the theoretical value of the power of the signal. (i.e. Keep between 4 & 5)
- Keep the signal attenuation at minimum using pot **P4**.
- Measure the signal power **Psni** on power meter.
- Increase the noise level by using **LEVEL** pot **P1.**
- Keep the attenuation pot **P4** to minimum position.
- Measure noise power **Pni** on power meter.
- Repeat the procedure by increasing the noise level in steps.
- Calculate signal to noise ratio at input of analog buffer by using formula

SNRi = (Psi / Pni)

- Keep the signal attenuation at minimum using pot **P4**.
- Measure the signal power **Psno** on power meter.
- Increase the noise level by using **LEVEL** pot **P1.**
- Keep the attenuation pot **P4** to minimum position.
- Measure noise power **Pno** on power meter.
- Repeat the procedure by increasing the noise level in steps.
- Calculate signal to noise ratio at input of analog buffer by using formula

SNRo = (Pso / Pno)

- Noise figure can be calculated by using following formula

NOISE FIGURE = (SNR) i/p / (SNR) o/p

Signal Power (Psni)	Noise Power (Pni)	Psi = (Psni-Pni)	(Snr) I/P
2.1	0.1	2.0	20
2.2	0.2	2.0	10
2.4	0.3	2.1	7
2.5	0.4	2.1	5.25
2.6	0.5	2.1	4.2

(Table given for reference only actual values may differ from above)

SIGNAL POWER (Psno)	NOISE POWER (Pno)	Pso = (Psno-Pno)	(SNR) o/p	Noise Figure
2.1	0.1	2.0	20	1
2.2	0.2	2.0	10	1
2.4	0.3	2.1	7	1
2.5	0.4	2.1	5.25	1
2.6	0.5	2.1	4.2	1

(Table given for reference only actual values may differ from above)

B. Input Signal: AM Signal

Similar observations and calculations can also be performed with AM signal as input signal using following formula

Noise power: just give the noise signal as input to multiplier and choose appropriate integrating time.

Signal power: same as above.

To calculate PSD: measure noise power by keeping the integrating time at max of 15s.

Bandwidth of the source is 2 MHz.

Single sided PSD = (noise power)/bandwidth

No = usually double sided PSD = single sided PSD/2

Signal to noise ratio is calculated using diff procedures for diff modulation techniques.

SNRi = modulated signal power / noise power in message bandwidth

Modulated signal power: same procedure as signal power

Noise power in message bandwidth: Nox 2x Receiver bandpass filter bandwidth

SNRo = demodulated signal power (without noise)/ noise power in demodulated signal

First measure demodulated signal power with noise

Then turn off noise. Measure signal power at output

Difference between the two gives noise power.

NOISE FIGURE = SNRi / SNRo.

NOTE: Power meter counts for first 15 seconds then shows stable output for 15 seconds, this cycle repeats continuously. All measurements should be done when display is stable.

NOTE: Make Sure PRBS is generate.

SWITCH FAULTS: -

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will open diode D1 across R58 & R59; this will increase ON time of astable multivibrator to about 52 seconds. Thus, time between successive readings get increases.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will open latch enable signal to U17 (pin11) from U11 (pin 8) the value controlled by counter won't get latched.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will open connection between transistor Q1 base & R55. So ramp generation will be lost. This will disable the counter and ADC operation will be affected.

EXPERIMENT
NO.4

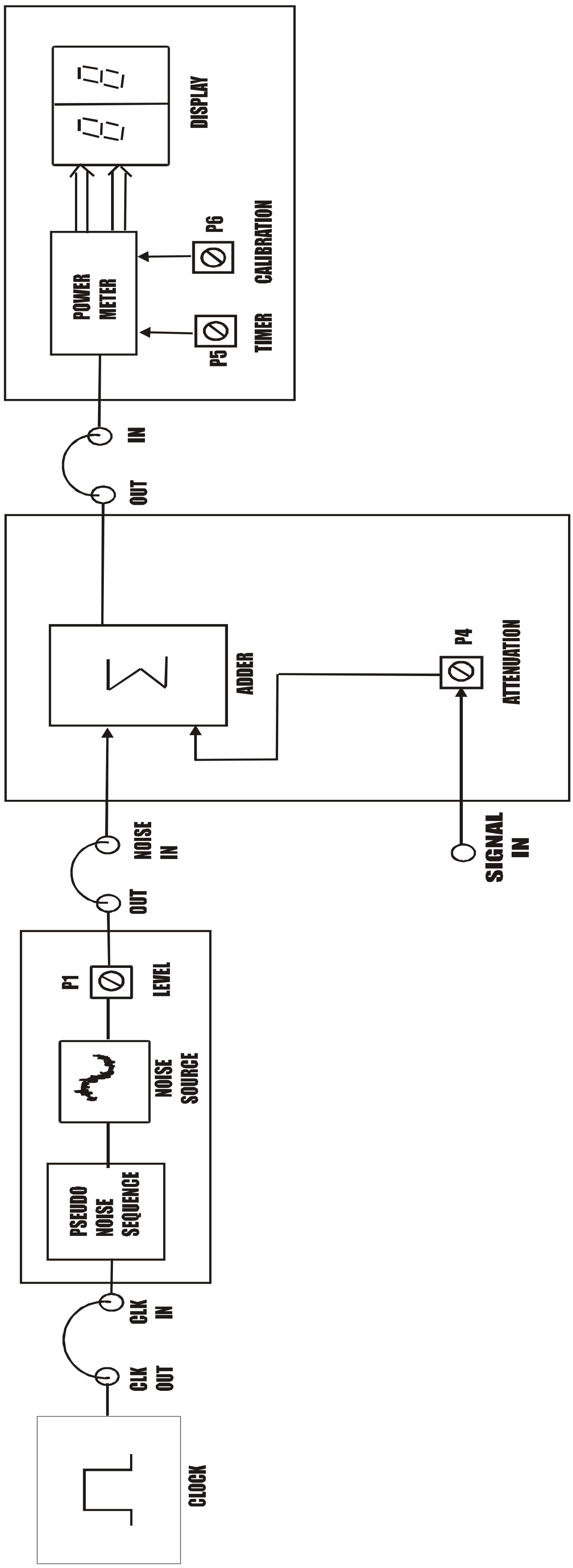


FIG. NO 4.1

FOR MEASUREMENT OF NOISE POWER SPECTRAL DENSITY

EXPERIMENT NO.4

NAME:-

To Study Measurement of Noise Power and Noise Power Spectral Density.

OBJECTIVES:-

To measure the power spectral density of PN sequence noise signal.

THEORY:-

Noise power: just give the noise signal as input to multiplier and choose appropriate integrating time.
Signal power: same as above.
To calculate PSD: measure noise power by keeping the integrating time at max of 15 seconds.
Bandwidth of the source is 2 MHz.
Single sided PSD = (noise power) / bandwidth
No = usually double sided PSD = single sided PSD/2
Signal to noise ratio is calculated using diff procedures for diff modulation techniques.

EQUIPMENTS: -

- ACL-NP.
- Power supply.
- 20MHz Dual trace oscilloscope.

NOTE: Keep all switch faults in OFF position.

PROCEDURE: -

- Connect the power supply with proper polarity to the kit **ACL-NP** and switch it **ON**.
- Ref to the fig. **4.1** & carry out the following connections and settings.
- Connect the **CLK OUT** post to **CLK IN** post of noise generator.
- Connect **OUT** post of noise generator to **NOISE IN** post of adder
- Keep the **LEVEL** pot **P1** of noise generator to maximum position.
- Keep the attenuation pot **P4** to minimum position.
- Connect **OUT** post of adder to **IN** post of power meter.
- Keep Timer pot **P5** to maximum position.

- Measure noise power **Pni** on power meter.
- **Single sided PSD = (noise power) / bandwidth.**
  No = single sided PSD / 2.

CALCULATIONS: -

Let Noise Voltage be 1Vpp
We will get Noise power on display as 0.5 Watt
Single sided PSD = Noise power / bandwidth
= 0.5 / 2 MHz
= 0.25×10^{-6} Watt/Hz
Noise Power Spectral Density = Single sided PSD / 2
= $0.25 \times 10^{-6} / 2$
= 0.125×10^{-6} Watt/Hz

NOTE: Power Meter Counts for First 15 Seconds Then Shows Stable Output For 15 Seconds, This Cycle Repeats Continuously. All Measurements Should Be Done When Display Is Stable.

NOTE: Make Sure PRBS is generate

SWITCH FAULTS: -

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will open diode D1 across R58 & R59; this will increase ON time of astable multivibrator to about 52 seconds. Thus, time between successive readings get increases.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will open latch enable signal to U17 (pin11) from U11 (pin 8) the value controlled by counter won't get latched.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will open connection between transistor Q1 base & R55. So ramp generation will be lost. This will disable the counter and ADC operation will be affected.

EXPERIMENT NO.5

EXPERIMENT NO.5

NAME: -

To Study Effect of Low Pass Filter on Noisy Signal.

OBJECTIVES:-

Observe the effect of low pass filter on noisy signal.

THEORY:-

Characteristics of low pass filter states that any signal with frequency component higher than its pass band frequency is rejected at the output of filter. Only frequency components lower or equal to the cut off frequency are passed through filter. In this kit we have low pass filter cut off frequency of 10KHz and maximum signal frequency onboard is also 10KHz, Generated noise signal may contain any frequency higher than 10KHz. Thus when we apply noise signal to the input of Low Pass Filter, only the signal frequency components are passed through the filter.

EQUIPMENTS:-

- ACL-NP.
- Power supply.
- 20MHz Dual trace oscilloscope.

NOTE: Keep all switch faults in OFF position.

PROCEDURE:-

- Connect the power supply with proper polarity to the kit **ACL-NP** and switch it **ON**.
- Ref to the fig. **5.1** & carry out the following connections and settings.
- Connect the **CLK OUT** post to **CLK IN** post of noise generator.
- Connect **OUT** post of noise generator to **NOISE IN** post of adder.
- Keep the **LEVEL** pot **P1** of noise generator to minimum position.
- Connect **OUT** post of function generator to **SIGNAL IN** post of adder.
- Keep the attenuation pot **P4** to minimum position.
- Keep freq. pot **P2** to minimum position i.e. at **1KHz.**
- Using pot **P3** Keep signal amplitude at **1Vpp**.
- Connect **OUT** post of adder to **IN** post of power meter.

- Keep Timer pot **P5** to maximum position.
- Measure signal power **Ps** on power meter.
- Increase the level of noise by varying the **LEVEL** pot **P1** of noise generator.
- Keep the attenuation pot **P4** to maximum position.
- Measure noise power **Pn** on power meter.
- Connect **OUT** post of adder to **IN** post of low pass filter.
- Connect **OUT** post of low pass filter to **IN** post of power meter.
- Measure signal power **Psf** on power meter.

NOTE: Power meter counts for first 15 seconds then shows stable output for 15 seconds, this cycle repeats continuously. All measurements should be done when display is stable.

NOTE: Make Sure PRBS is generate.

SWITCH FAULTS: -

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will open connection between U4 (pin 12) & U5A (pin 1) of noise generator section, this will reduce the amplitude of generated noise. This will also reduce the power of noise signal.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will open diode D1 across R58 & R59; this will increase ON time of astable multivibrator to about 52 seconds. Thus, time between successive readings get increases.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will open latch enable signal to U17 (pin11) from U11 (pin 8) the value controlled by counter won't get latched.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will open connection between transistor Q1 base & R55. So ramp generation will be lost. This will disable the counter and ADC operation will be affected.

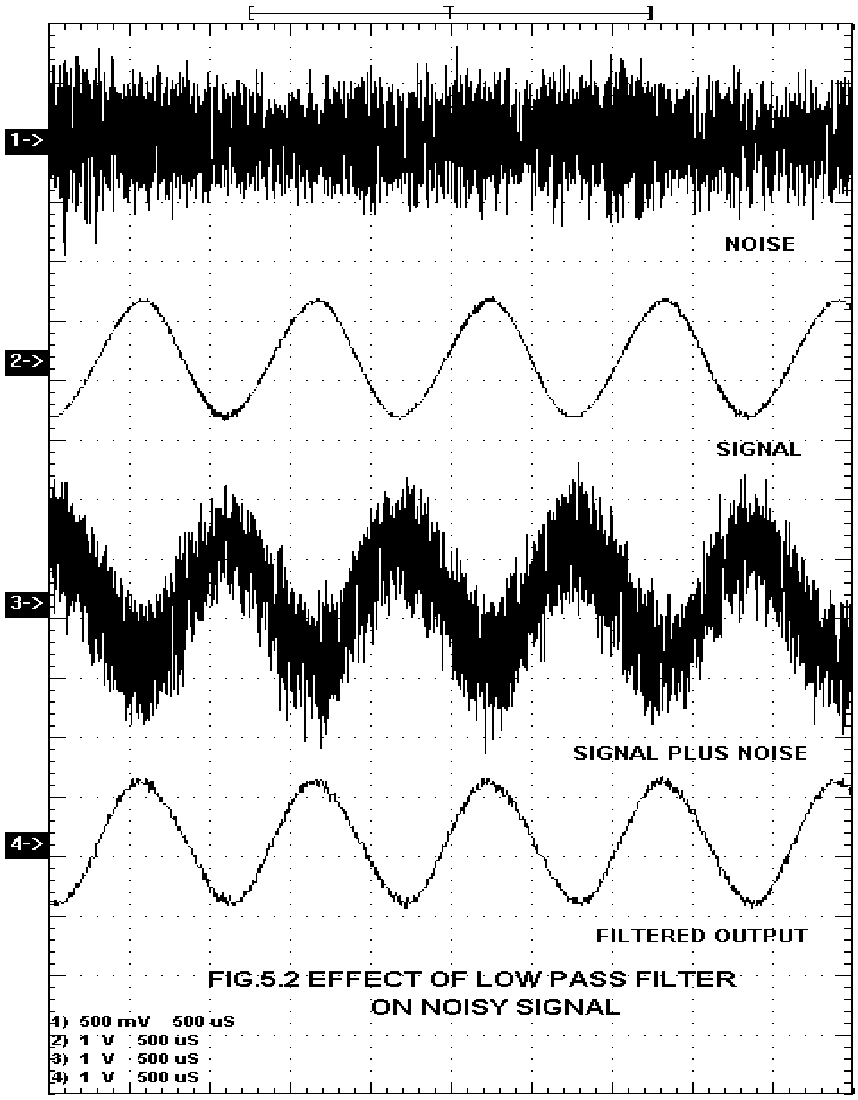


FIG.5.2 EFFECT OF LOW PASS FILTER ON NOISY SIGNAL